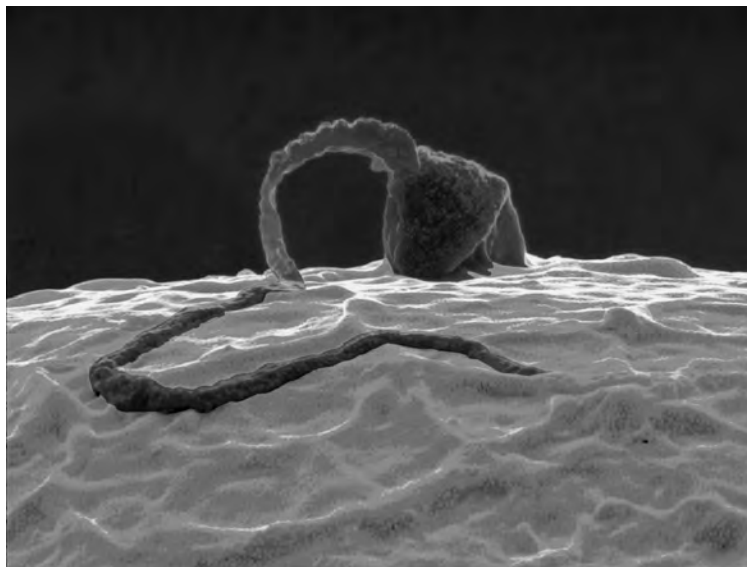


- 4 The photograph shows a sperm cell beginning to enter an egg cell.



(Source: © THIERRY BERROD, MONA LISA PRODUCTION / SCIENCE PHOTO LIBRARY)

- (a) A sperm cell swims by rotating its flagellum.

Which of the following statements about sperm cells are correct?

1. sperm cells are diploid
2. the flagellum rotates using energy
3. sperm cells have a streamlined shape to increase resistance

(1)

- ☐ **A** 1, 2 and 3
- ☐ **B** 1 and 2 only
- ☐ **C** 2 and 3 only
- ☐ **D** 2 only



(b) Describe how the nucleus of a sperm cell enters an egg cell.

(3)

(c) Cortical granules in the egg cell contain enzymes.

These enzymes prevent more than one sperm entering the egg cell.

(i) Describe the role of the rough endoplasmic reticulum and the Golgi apparatus in the production of these enzymes.

(3)



- (ii) Suggest how these enzymes will prevent more than one sperm nucleus entering the egg cell.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 4 = 10 marks)

